

FAR WESTERN UNIVERSITY
Supplementary Examination-2079
Engineering Survey II (GE245)

Faculty: Engineering (Civil)

Level: Undergraduate

Semester: Fourth

Full Marks: 100

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate *full marks*.

Group – A

Attempt any five questions.

(Short answer questions)

5×8=40

1. Explain in brief field procedure of traverse survey between two known stations.
2. A tachometric survey was done to find the gradient between X and Y. Tacheometer consist of an analytic lens was used and following observations were made from Station R on vertical staff.

Inst. Station	Staff Station	Stadia Readings(m)	Vertical Angle	Bearing
R	X	0.915, 1.750, 2.585	+ 15°	345°
	Y	0.760, 2.240, 3.715	+ 10°	75°

3. How can you measure the horizontal distance and elevation of an inaccessible object when the instrument positions are at very different levels; instrument stations and the elevated object are in the same vertical plane.
4. Explain degree of curve with neat sketch. Derive the formula of Tangential angle, $\delta = \frac{90^\circ \times C}{\pi R}$ and deflection angle $\Delta_n = \delta_1 + \delta_2 + \delta_3 + \dots + \delta_n$.
5. What are the different methods of interpolating contours? Explain them with appropriate examples and sketch.
6. Explain any one method of analytical resection to calculate the position of instatement station.

Explain

Group – B

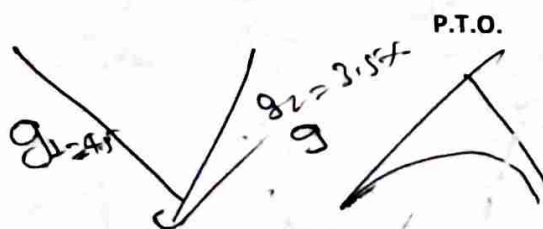
Attempt any four questions.

(Long answer questions)

4×15=60

7. a. What is GPS? Write the components and working principle of GPS.
- b. Prepare a table giving all necessary data for setting out a vertical curve. In a road alignment a grade of - 4.5% followed by + 3.5% R.L. of I.P. = 1000 m chainage of IP = 1500 m connect the two grade by a parabolic curve 200 m long. Take peg interval = 20 m. (10)

Curve



$\cos R = \frac{x}{R}$

0.2

8. a. Determine the expression for the scale of vertical photograph. (5)

b. A traverse ABCDA was conducted and the following data were obtained. It was required to connect the mid-point E of CD to the mid-point F of AB. Find the length and bearing of line joining mid points E and F. (10)

Line	Length in (m)	Bearing
AB	610.00	N 80°10' E
BC	510.00	N 13°00' E
CD	1130.00	S 80°10' W
DA	450.00	S 15°30' E

9. a. Define sounding, and state the equipment and methods used for locating and making soundings. (7)

b. Explain the principle of Tachometric survey and also derive the formula to determine the horizontal distance and RL of the object with respect to instrument station when the staff is held in vertical. (8)

10. a. Describe contour interval and horizontal equivalent. Explain characteristics of contour with neat sketches. (7)

b. The co-ordinates of S and A are (1309.12 m E, 1170.50 m N) and (1525.43 m E and 956.87 m N) respectively. Calculate the co-ordinates of point B which has been by intersection from stations S and A observing the following angles. $\angle BSA = 85^\circ 38' 49''$ and $\angle SAB = 55^\circ 50' 33''$. (8)

11. Write short notes any three (3 x 5 = 15)

- Components of GIS and its application.
- Relief displacement and its expression.
- Accuracy parameters of horizontal and vertical control of traverse.
- Application of GIS in Civil Engineering.

Handwritten signatures and marks are present below the questions, including a circled '20' and various scribbles.

(2019) Regular

Full Marks: 100
Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt any five questions.

Group - A
(Short Questions)

5×8=40

1. Define closed loop and closed link traverse. Explain in brief the field measurements required in theodolite traversing with accuracy parameters (or specifications). (8)

2. For a closed traverse ABCDA, compute the missing data. (8)

Line	Length(m)	Bearing
AB	100.00	N 45° 30' W
BC	605.00	N 5° 30' E
CD	95.00	N 88° 20' E
DA	?	?

3. Explain the principle of Tachometric survey and also derive the formula to determine the horizontal distance and RL of the object with respect to instrument station when the staff is held in vertical. (8)

4. Calculate the distance and gradient between stations A and B from the given data which are observed by a tachometer from station R. Staff was vertically held at A and subtense bar at B. The subtended angle between the instrument and 2m long subtense bar was 00°42'15". (8)

Instrument Station	Sighted to	Bearing	Zenith angle	Staff readings (m)	Subtense bar height
R	A	345°00'	96°30'	0.650, 1.250, 1.850	-
R	B	225°00'	85°00'	-	1.180 (m)

5. Explain the direct method of contouring and write the uses of contour map. (8)

6. Prepare a table giving all necessary data for setting out a vertical curve. In a road alignment a grade of - 4.5% followed by + 3.5%, R.L. of I.P.: 1000 m chainage of IP = 1500 m connect the two grade by a parabolic curve 200 m long. Take peg interval: 20 m. (8)

PTO

Attempt any four questions. (Long Questions)

7. (5) a) With the help of neat sketch explain with expression the relief displacement of a vertical photograph.

b) Stations A, B and C have the following respective coordinates (2876.24 mE, 8754.11 mN), (3310.80 mE, 7997.25 mN) and (2959.39 mE, 7487.09 mN) respectively. Station 'O' was established and following observations were recorded by a theodolite.

Pointing towards	Horizontal Circle Reading
A	90°00'00"
B	230°58'51"
C	313°17'05"
A	90°00'30"

Determine the co-ordinate of resection point 'O'.

(10)

8. (7) a) What is GPS? Write the components and working principle of GPS.

b) It is required to determine the height (clear) of a Flood light tower in an arena by using a transit-theodolite and for this zenith angles observation taken at 5 m and 2 m height on a target vane held on the plinth level of tower were 87°45' and 88°30' respectively. From the same instrument, zenith angle observed at top of the tower was found as 67°45'. If the RL of the instrument axis was 1200.00 m, Calculate the clear height (plinth to top) of the tower.

(8)

9. (7) a) Compute the data for setting out a simple circular curve by Rankine's deflection angle method from the following information:

Angle of intersection = 145°00'
Chainage of point of intersection = 1580 m
Degree of curve = 5°
Least count of the theodolite = 10"
Peg interval = 30 m

b) Define vertical and horizontal control. Explain the methods of locating sounding points in hydrographic surveying. (8)

10. (5) a) What is contour interval? Explain the factors that affect the contour interval.

b) Balance the following coordinates and compute the total coordinates for a link traverse XABCY using Bowditch's rule. The given coordinates of X and Y are (1877.51 m N, 1290.20 m E) and (1626.50 m N, 1578.87 m E) respectively. If permissible closing error is 1:500, justify your assessment. Other observed data are as follows: (10)

Line	XA	AB	BC	CY
Length(m)	120	111.50	132.40	97.60
Bearing	135°00'	119°30'	175°00'	77°30'

11. Write short notes on: (any three)

(3x5=15)

- Component of GIS and its application.
- Application of remote sensing in civil engineering.
- Expression for the scale of vertical photograph.
- Expression for deflection angle of transition curve.

15 15 15
45

FAR WESTERN UNIVERSITY
Supplementary Examination-2079

Faculty: Engineering (Civil)
Level: Undergraduate
Semester: Fourth
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Hydraulics (CE247)

Full Marks: 100

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt any five questions.

Group - A

(Short answer questions)

5×8=40

1. a) Oil of density 750 kg/m³ and dynamic viscosity 0.24 poise flows through 60mm diameter pipe of length 1000m at the rate of 0.25 lps. Determine (i) pressure gradient, (ii) wall shear stress. [4]
b) Show mathematically how you can relate the centerline and local velocities for a turbulent flow in pipes with supporting sketches. [4]
2. A 180mm diameter siphon connects two reservoirs having a difference in elevation of 16m. The length of siphon is 450m and the summit is 4.5m above the upper reservoir level. The length of the pipe from upper reservoir to the summit is 90m. Determine the discharge through the siphon and pressure at the summit. Consider all losses. Take $f = 0.025$. [4]
3. a) A steel pipe is connected directly to an oil storage reservoir. The valve provided at the end of the pipe is suddenly closed to stop the flow oil. Explain with necessary figures the variation of pressure in the mid-section of the steel pipe. [5]
b) Differentiate gradually, rapidly and spatially varied flow with neat sketches and examples. [3]
4. A trapezoidal channel with bottom width 2m and side slope 2V:1H, flows with a depth of 0.8m in a slope of 0.001, compute bed shear stress, shear velocity, Chezy's C and friction factor f. [8]
5. What do you mean by energy slope? An open channel of the most economic cross section, having the form of half hexagon with horizontal bottom is required to carry a maximum discharge of 20.2 m³/s of water. The slope of the channel bottom is 1 in 2500. Taking Manning's $n = 0.015$, determine the dimensions of the cross sections. [8]
6. a) Plot the possible types of GVF profiles for combined slopes with steep slope followed by mild slope, given that the mild slope is provided with a sluice gate. [3]
b) How are sequent depths related? Derive the relation with necessary assumptions. [5]

P.T.O.

Group - B
Attempt any four questions. (Long answer questions)

4×15=60

7. a) The water surface of reservoir B is 10m below the water surface in reservoir A. The reservoirs are connected by horizontal compound pipe system 1 and 2. The pipe details are

Pipe	Length (m)	Diameter (m)	f
1	250	0.25	0.007
2	450	0.35	0.008

- i) Compute all the head losses [7+5]
ii) Sketch EGL and HGL
- b) What is water hammer phenomena? Also point out its effects and remedial measures. [1+2]
8. a) Reservoir A with water surface elevation 120m is connected to reservoir B and C having water surface elevations of 70m and 50m respectively. A pipeline 170mm diameter and 450m long connects reservoir A to junction D. Reservoir B and C are connected to junction by 80mm diameter 110m long and 90mm diameter 220m long pipeline respectively. Assuming friction factor $f = 0.035$ for all pipes, estimate the discharge through all pipes. Neglect minor head losses. [12]
- b) Determine the geometric properties of a sewer line made up of precast RCC pipe with flow depth smaller than its radius. [3]
9. The depth of uniform flow in a rectangular channel 5m wide ($n = 0.014$, bed slope = 1 in 30) is 0.54m. A low dam raises the water to a depth of 3.1m. Find whether a hydraulic jump takes place and if so at what distance upstream of the dam. Use direct step method. [15]
10. a) Water flows through a rectangular channel, 3m wide, with a velocity of 1.25m/s. The depth of flow is 0.5m. Compute (i) specific energy, (ii) specific force, (iii) alternate depth, (iv) conjugate depth. [8]
b) A rectangular channel contracts from a width of 3m to 2.5m in a short smooth transition section. If the discharge is 6.5 m³/s, and the u/s depth is 0.4m, determine the d/s depth. [7]
11. Write short notes on:
i) Oscillation of liquid in a U-tube [5+5+5]
ii) Specific energy diagram
iii) Loss of energy in hydraulic jump in terms of pre-jump and post-jump velocities

$$3 + 3 + 15 + 5 + 3 + 4 = 43$$

6, 18, 4

15

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FAIR WESTERN UNIVERSITY
Semester End Examination-2079
Hydraulics (CE 247)

Faculty: Engineering (Civil)
Level: Undergraduate
Semester: Fourth

Full Marks: 100
Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Group - A

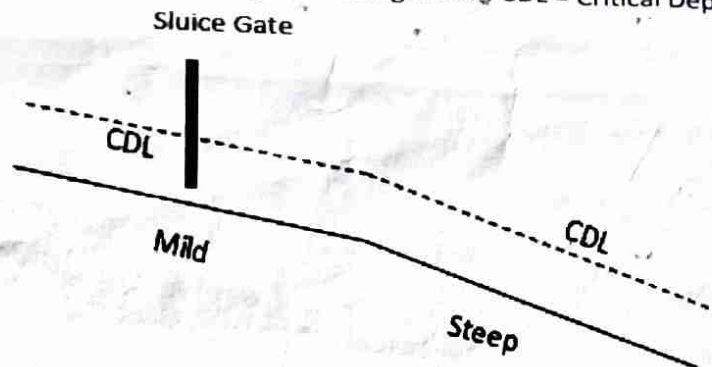
5×8=40

Attempt any five questions.

(Short Questions)

1. A CI water pipe of 0.5m diameter was originally designed to carry a discharge of 250lps in a water supply pipeline of Mahendranagar city. After 10 years of use, it was found out that the pipeline required 30% more power to carry the same discharge. Determine the corresponding roughness of pipe and rate of change in roughness if $\text{power} \propto \text{friction factor}$. Take $\nu_{\text{water}} = 0.014$ stokes. [6+2]
2. a) Calculate the diameter of a single pipe which can replace a series combination of three pipes of lengths 1000m, 400m and 600m and of diameters 0.5m, 0.4m and 0.3m respectively. [3]
b) Show mathematically how you can relate the maximum and point velocities for a turbulent flow in pipes with supporting sketches. [5]
3. a) A 800m long penstock pipe supplying water to turbine is connected directly to a reservoir of a storage type hydropower project. The valve connecting to the turbine was forced to close in 1.5 seconds due to sudden damage in the transmission line. Explain with necessary figures the variation of pressure in the penstock pipe at 400m d/s of the reservoir. Take $c = 1000\text{m/s}$ [5]
b) Identify the geometric properties of a sewer line made up of hume pipe with flow depth greater than its radius. [3]
4. a) Find the ratio of discharges in concrete lined rectangular and triangular channels, if the rectangular channel has flow depth equal to its width and the triangular channel has an apex angle of 90° , such that the flow areas are same for both the channels. [5]
b) Calculate the Chezy's coefficient for a triangular channel with an apex angle of 60° carrying a discharge of $1\text{ m}^3/\text{s}$ at a depth of 0.75m and laid at a slope of 0.009. [3]
5. A trapezoidal channel 2.5m wide and having side slopes 1:1 is carrying a discharge of $1.1\text{ m}^3/\text{s}$ at a specific energy of 0.6m. What would be the corresponding bed slopes required to sustain the uniform flow if $n = 0.015$. Also calculate the minimum specific energy required to carry this discharge. [8]

a) Plot the possible types of GVF profiles for given arrangement. CDL = Critical Depth Line. [3]



b) How are sequent depths related? Derive the relation with necessary assumptions. [5]

Group - B

Attempt any four questions. (Long Questions)

4×15=60

7. a) A 180m long pipeline ABC is laid on an upward slope of 1 in 40. The pipe details are

Section	Length (m)	Diameter (m)
AB	100	0.1
BC	80	0.2

The pipe suddenly enlarges at B. A discharge of $0.2 \text{ m}^3/\text{s}$ is pumped into the pipe at its lower end at A and is discharged into a closed tank at C. The pressure at supply end is 200 kN/m^2 . $f = 0.008$

i) What is the pressure at C?

ii) Draw HGL and EGL.

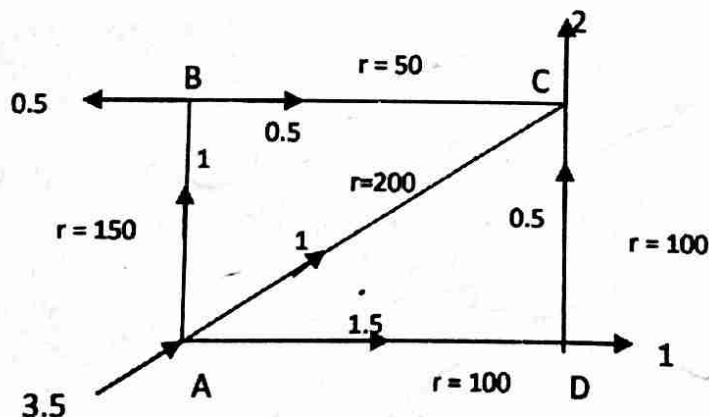
[6+6]

b) What is water hammer phenomena? Also point out its effects and remedial measures.

[1+2]

8. A water supply distribution network in Terai is as shown in the figure below. The numbers and arrows indicate initial assumptions for discharge and flow direction for each pipeline respectively. Perform two iterations to calculate the corrected discharge in the pipe network. If its inlet is supplied by a 20m high overhead tank, calculate pressure at points B, C and D neglecting velocity heads. State necessary assumptions clearly.

[10+5]



9. A 5m wide long rectangular channel has a vertical sluice gate with an opening of 0.7m producing a downstream jet depth of 0.38m supplying a steady discharge of $20 \text{ m}^3/\text{s}$. Assuming that the flow downstream of the gate eventually returns to the uniform flow depth of 2.5m,

$$n = 0.02 \quad S_0 = 0.01$$

a) Verify that hydraulic jump occurs.

b) What distance d/s of the vena contracta does the hydraulic jump occur? Use direct step method and solve in two steps.

c) Will the hydraulic jump be affected in any ways if the uniform flow depth changes to 3m? [4+8+3]

10. A discharge of $2.5 \text{ m}^3/\text{s}$ flows through a 2.2 m wide rectangular channel at a depth of 1 m . For the following changes in the channel section, check if critical flow occurs at a section with;

- Constriction in channel with width reduction to 1.5 m
- Introduction of a smooth hump 0.2 m high.
- Combination of hump and width constriction stated above.

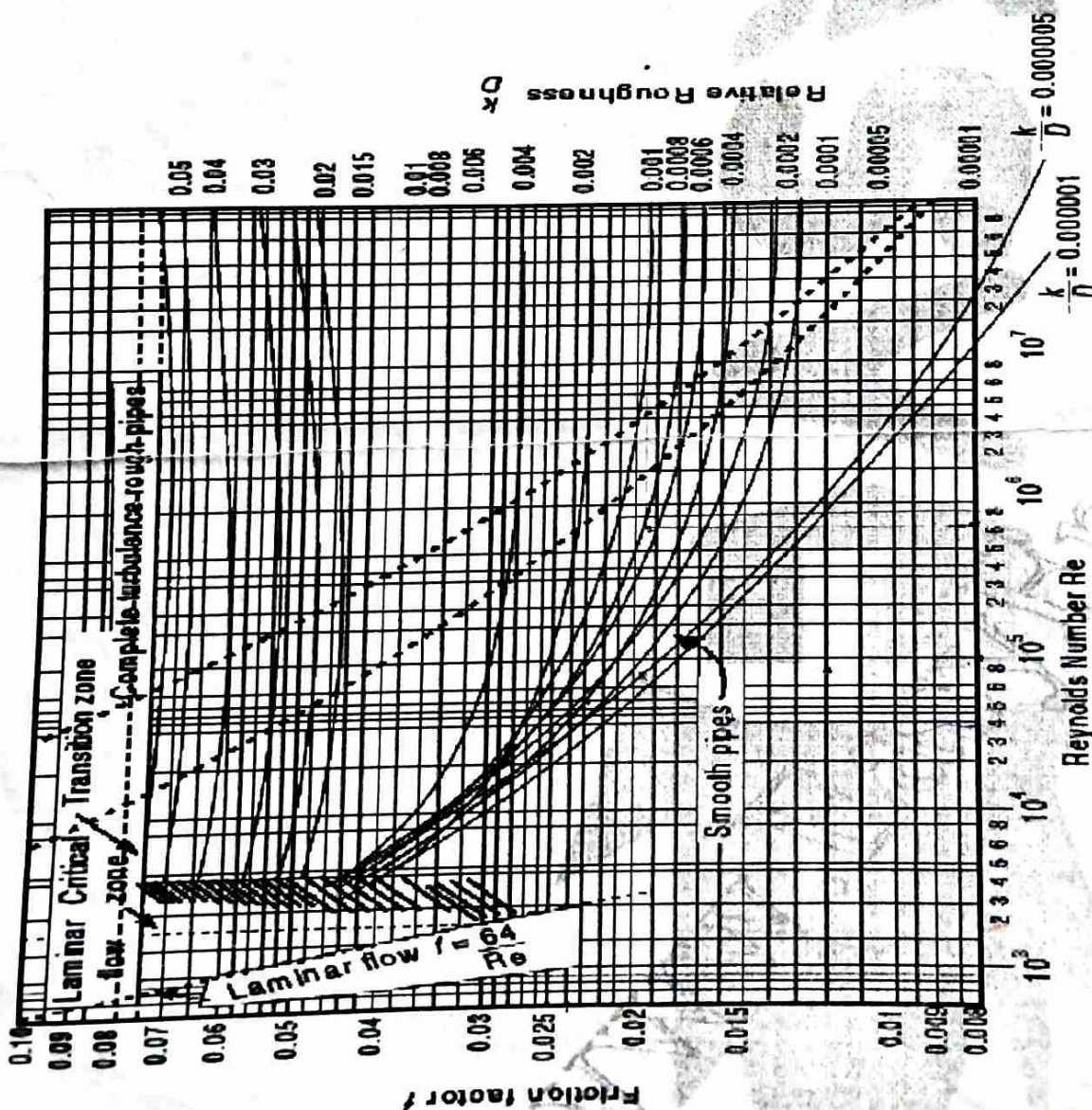
Will case (c) cause increase of u/s depth? If so, to what extent?

[4+4+3]

11. Write short notes on:

- Velocity distribution in open channels
- Application of momentum principle in open channel flow
- Loss of energy in hydraulic jump in terms of pre-jump and post-jump velocities

{ [5+5+5] }



FAR WESTERN UNIVERSITY
Semester End Examination-2079
Building Technology (AR 246)

Group -B

Faculty: Engineering (Civil)
Level: Undergraduate
Semester: Fourth

Full Marks: 100
Time: 3hrs.

Attempt any Four Questions.

4x15=60

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Group - A

5x8=40

Attempt any five questions.

1. Explain the process of setting layout plan of a building with necessary sketches.
2. What are the remedial precautions to be undertaken to prevent unequal settlement of the existing foundation? What type of foundation will you recommend for the foundation on black cotton soils? Give your justification.
3. A stair hall measure 4m x4m with floor height 3.3m. Design a suitable stair for the building with necessary sketches.
4. In your opinion which type of door is suitable for residential building & why? Show flooring details of a ground floor that has a connection with external wall and floor finish with mosaic tiles.
5. Due to natural disaster one footing of a building is completely damaged. In this situation which type of solution do you apply for this building & how it can be implemented? Also write merits and demerits of the proposed solution.
6. What do you mean by shrinkage cracks? Explain the factors that are adopted for preventing shrinkage cracks.

7. a) Differentiate pile foundation, pier foundation & caisson foundations.(8)
b) Define stone masonry with its types. Draw typical sketches to illustrate them. (7)
8. a) Define the term bond. What are the types of bond used in brick masonry? Explain any four of them with neat sketches. (8)
b) State need of retrofitting. What are the factors that should be considered in order to decide whether to retrofit or not? (7)
9. a) Write the different wiring systems in common uses with their advantages. (8)
b) What is septic tank? Explain the construction details of a septic tank with a neat sketch. (7)
10. a) Differentiate between king post and queen post roof truss with necessary sketches.(7)
b) Define joints with their different types. Where and how construction joints are provided for RCC beam, column and slab? (8)
11. a) Why earthing is required for building? How to control the crime in public place by using close-circuit television? (7)
b) Why non-destructive test is used in building? Which method do you apply for non-destructive test on structures? (8)

FAR WESTERN UNIVERSITY
Semester End Examination-2079
Communication English II (SH 241)

Faculty: Engineering (Civil)
Level: Undergraduate
Semester: Fourth

Full Marks: 100
Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Group – A

Attempt any five questions.

(Short Questions)

5×8=40

1. Discuss the strategy for effective listening skills?
2. Explain the ways to improve your reading skills?
3. Discuss the principles of effective speaking?
4. The tuition costs for the children of employees at ABC College will henceforth be covered by the college to the tune of 50%. Create a memo to explain this policy.
5. A manufacturer in the food-processing sector with rapid growth, Sammy Agro Industry, has openings in its training program. Only individuals with exceptional communication skills who are highly motivated, dynamic, and results-oriented need apply. Performance determines how quickly one can rise to management roles. Create a professional-looking job application. Send your resume to Mahendranagar, PO Box 976, Personnel Manager.
6. For your research project, pick a subject that interests you. After choosing your research topic, compose the introduction and methodology for the study.

Group – B

4 x 15 = 60

Attempt any four questions.

7. Consider yourself a senior manager at Hill Resort who is tasked with reviewing the 15 applicants for the position of front desk receptionist. Encourage those who failed in a letter without apologizing and reassure them that their faculty is strong. Use complete block format.
8. Your business, Bright Star Logistics, which was founded in Mahendranagar, is looking at potential locations in and around the Dhangadhi Industrial Area for a new branch. You are required to submit a report as the company's director. Create your own title for a feasibility report on the same topic.
9. Prepare a report on the "Importance of hybrid vehicles during the gasoline scarcity" with a cover page, title page, acknowledgements, conclusion, and recommendations.
10. The General Manager of your organization is concerned to discover that an increasing trend among employees is "to take time off from work without asking for permission." He requested that you, the personnel manager, investigate the root reasons and make recommendations for corrective action. Write a memo report detailing this pattern, the issues that are contributing to it, and the solutions.
11. Draft a brief proposal to the dean's office on "Installing air conditioners in classrooms to avoid sweltering heat and ensure the effective teaching-learning environment" with objectives and procedures.

FAR WESTERN UNIVERSITY
Supplementary Examination- 2079
Probability and Experimental Design (SH244)

Faculty: Engineering (Civil)

Level: Undergraduate

Semester: Fourth

Full Marks: 100

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Group – A

Attempt any five questions. (Short answer questions)

5×8=40

1. With example describe how can you relate your statistical knowledge in your related field?
A purchasing agent obtained samples from two suppliers and he had tested the samples in his own laboratory for length of life with the following results

Length of life(hr)	700-900	900-1100	1100-1300	1300-1500
Sample A	10	16	26	8
Sample B	3	24	12	3

Decide which company's bulb do you want prefer?

(2+6=8)

2. Define conditional probability and Bayes theorem. In a factory A produces 60% of total output and B produces 40% of total output. Further 8% of output of machine A and 6% of outputs of machine B are defective. In a day's production these two machines produce 10000 items. If an output is selected at random is defective, what is the probability that it was produced from machine B?
(2+6=8)
3. Define binomial experiment and binomial random variable with example. Suppose two coins are tossed together and X denotes number of head, find the probability distribution of X. Also calculate the probability of at most two heads.
(2+4+2=8)
4. The daily wages of 1000 workers is normally distributed around a mean of Rs. 70 and with standard deviation of Rs.5. estimate the number of workers whose daily wages will be (i) between Rs. 70 and Rs. 72 (ii) between Rs. 69 to Rs. 72 (iii) more than Rs. 75 (iv) less than Rs. 63.
(2+2+2+2=8)
5. The joint probability density function for two random variables X and Y is given by

$$f(x, y) = \begin{cases} c(x+y)^2, & \text{for } 0 < x < 1 \text{ and } 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$
 find (i) value of c
 (ii) $P(0.2 < X < 0.5 \cap 0.4 < Y < 1)$ (iii) the marginal density function of X.
(2+3+3=8)
6. A population consist of five numbers 1, 3, 5, 7 and 9. Enumerate all possible samples of size 2 which can be drawn from the population without replacement. Show that the mean of the sample mean is equal to the population mean.
(4+4=8)

P.T.O.

Group - B

Attempt any four questions. (Long answer questions)

4=15=60

7. Define the meaning of estimation and discuss with particular example that how can you apply the concept of estimation in your related field. The two independent samples of sizes 80 and 35 drawn from two normal populations have means 50 and 45 with variances 16 and 25 respectively. Obtain 95% confident interval for difference of population means $(\mu_1 - \mu_2)$. (2+5+8=15)
8. Discuss the meaning of large and small sample in any statistical investigation and also discuss the meaning of hypothesis, hypothesis testing and its importance in research. A company claims that its light bulbs has mean lifetime 15000 hours of continuous use with s.d of 120 hours. However there has been a against claim of customer that a sample of 60 bulbs has mean lifetime of 13000 hours of continuous use. Can you justify this claim at 0.02? (2+5+8=15)
9. Define and differentiate between correlation and regression and discuss their different types with example. Also discuss their importance in engineering sector. The shear resistance of soil, τ KN m² is determined by measurement as function of the normal stress X KN m² is as follow

X	10	11	12	13	14	15	16	17
Y	14.08	15.57	16.94	17.68	18.49	19.55	20.68	21.72

Estimate the approximate shear resistance of soil when normal stress is 21 KN m². (4+3+8=15)

10. Discuss the meaning and importance of analysis of variance (ANOVA) and also discuss why it is different from Z-Test and T-test. Three training methods were compared to see if they led to greater productivity after training. The following are productivity measures for individuals trained by each method

Method 1	45	40	50	39	52	44
Method 2	59	43	47	51	39	49
Method 3	41	37	43	40	52	39

Perform an analysis of variance to test at 0.05 level of significance whether the three training method lead to different levels of productivity? (2+1+12=15)

11. Discuss the meaning and importance of paired t-test. Blood pressures of 5 pressure patients were tested before and after prescribing certain medicine. Test at 5% that whether the medicine is effective or not. (3+12=15)

patient	1	2	3	4	5
BP before medicine	135	140	143	143	155
BP after medicine	120	131	128	122	150

FAR WESTERN UNIVERSITY
Semester End Examination-2079
Probability and Experimental Design (SH 244)

Faculty: Engineering (Civil)
Level: Undergraduate
Semester: Fourth

Full Marks: 100
Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Group – A

Attempt any five questions. (Short Questions)

5×8=40

1. Differentiate between descriptive statistics and inferential statistics and also describe their tools with examples. An analysis of monthly wages paid to workers in two firms A and B gives following results:

	Firm A	Firm B
No. of workers	550	650
Average wage(Rs)	190	185
Variance of distribution of wage	81	100

(a) In which firm A or B, is there greater variability in individual wages?

(b) Calculate the average monthly wages and variance of distribution of wages of all workers in both firms.

(2+6=8)

$(\bar{x}_{12})$, (s_{12})

2. Do you think that there is any difference between conditional probability and Bayes theorem? If any describe with example. In a certain campus, 25% of boys and 10% of the girls are studying mathematics. The girls constitute 60% of students'. If a student is selected at random and is studying mathematics, what is the probability that the student is girl. (2+6=8)

3. Define the discrete random variable, discrete probability distribution and probability mass function with example. Two fair dice are rolled and X denotes the sum of two numbers that appear on top. Find the probability distribution of X. Also find the mean variance of the distribution. (2+5+1=8)

4. Define continuous random variable and its probability density function with example. The marks obtained by 300 students in mathematics are normally distributed with mean score 68 and standard deviation 3. How many students have obtained marks (a) greater than 72 (b) less or equal to 64 (c) Between 65 and 71 inclusive.

5. Define two dimensional random variable and its probability mass and density function. If two random variable have joint density

$$f(x, y) = \begin{cases} \frac{6}{5} (x + y^2) & \text{for } 0 < x < 1, 0 < y < 1 \\ 0 & \text{else} \end{cases}$$

(a) Find the marginal function of X and Y

(b) $P(0.2 < x < 0.5)$ and $P(0.4 < y < 0.6)$

6. With example define estimate, estimator and estimation. Also describe the importance of estimation in engineering field. A researcher wants to estimate universe mean by using sampling technique. What should be the sample size when the permissible error between parameter value and sample statistic in 95% of chance will not be more than 1.5 and population s. d. is 15.

(24)

$\bar{x} = \mu = 1.5$
 $\frac{60}{15} = 4$
 $n = 16$

24

8

32

Q. Do you think that estimation and hypothesis testing are essential tools of making inferences about population? Express your views with suitable examples. Two groups, each made of 9 students, were selected randomly. A course of mathematics was taught to first group by discovery method and second group of 9 student was taught same course by explanation method. At the end of the course same test was given two groups. The first group made a mean marks 35.22 with a variance of 24.445, while second group made a mean marks 31.56 with a variance of 20.0275. Is there a significance difference between the effectiveness of two methods? (3+12=15)

8. Why do we need ANOVA? Explain the weakness of Z-test and T-test. Explain briefly about one way ANOVA with its application. The following are the breaking strength of three different brands of cables

Brand	Breaking strength					
A	40	30	50	60	30	----
B	60	40	55	65	----	----
C	60	50	70	65	75	40

Construct ANOVA table and test for the equality of average breaking strength of cables at 5%. (1+1+2+11=15)

9. (a) Differentiate between correlation and regression with example. The marks secured by 10 students in Statistics (X) and research methodology (Y) are

Serial no.	1	2	3	4	5	6	7	8	9	10
X	52	50	55	50	60	70	72	78	80	63
Y	20	25	60	45	80	25	55	65	25	75

Calculate the rank correlation coefficient and interpret the result. (8)

10. (a) Define parameter and statistics. Write the difference between point estimation and interval estimation with examples. A sample of 500 bulbs of company showed an average life 1400 hours with s.d. of 30 hours. Obtain 95% and 99% confident interval for population mean. (8)

(b) Why normal approximation to binomial distribution is necessary? Discuss your views with examples. If 20% of the memory chips made in a certain plant are defective, what the probability that in a lot of 100 randomly chosen for inspection (I) at most 15 will be defective (II) between 10 and 15 inclusive. (2+5=7)

11. (a) When do you use the paired t-test and how to apply it. A certain stimulus administered to each 12 patients resulted in following increase of blood pressure: 5, 2, 8, -1, 3, 0, -2, 1, 5, 0, 4, 6. Can it be concluded that the stimulus effective in increase of blood pressure. (8)

(b) Define negative binomial distribution and state the condition of applicability for it. Write the similarities and difference between binomial and negative binomial distribution. A student has taken 5 answers multiple choice examinations orally. He continues to answer questions until he get 3 correct answers. What is the probability that he gets them at most 7 questions if he guesses at each answer? (2+5=7)

FAR WESTERN UNIVERSITY
Semester End Examination-2079
Structural Analysis I (CE 243)

Faculty: Engineering (Civil)
Level: Undergraduate
Semester: Fourth

Full Marks: 100
Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate *full marks*.

32

Group – A

Attempt any five questions.

(Short Questions)

24
5×8=40

1. State principle of superposition with one example and give reasons why the principle of superposition couldn't be applied for nonlinear structures. ✓ 8
2. What do you mean by virtual work method and real work method? Which of the methods is superior? Give some examples to justify your answer. ✓ 8
3. Determine the horizontal displacement of the Joint C of the portal frame as shown in the fig. 1 by strain energy method. EI is Constant throughout. ✓ 8

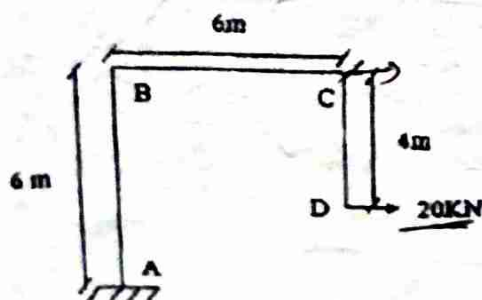


Fig. 1

$W = \frac{1}{2} P \Delta$

4. Draw the influence line diagram for U_3 , U_2 , L_2 and U_1 of the truss shown in the figure 2. Consider a unit load moves in the lower chord from L_0 to L_6 .

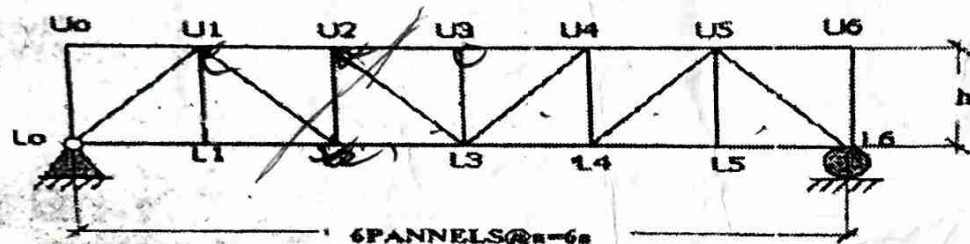


Fig. 2

5. A three-hinged parabolic arch has a span of 40 m and rise of 8 m. Determine the bending moment, normal thrust and radial shear at 15 m from the left support, if the arch is subjected to the uniformly distributed load of 25 kN/m over the left half portion and a concentrated load of 60 kN at 30 m from the left springing.



6. Using Macaulay's method, find the deflection at the mid span and slope at the right support of a simply supported beam of span 8 m subjected to three point loads of 20 kN at equal interval between the supports.

wl^2



Group - B

Attempt any four questions.

(Long Questions)

7. Derive Conjugate beam method from moment area method and determine the deflection at B and D using moment area method or conjugate beam method for the beam shown in the fig3. Assume $E = 12600 \text{ kN/m}^2$ and beam has width of 350 mm and depth of 500 mm.

$15 + 8 = 24$

$4 \times 15 = 60$

wl^2

$\frac{wl^3}{over}$

$\frac{wl^2}{48EI}$

$\frac{wl^2}{48EI}$

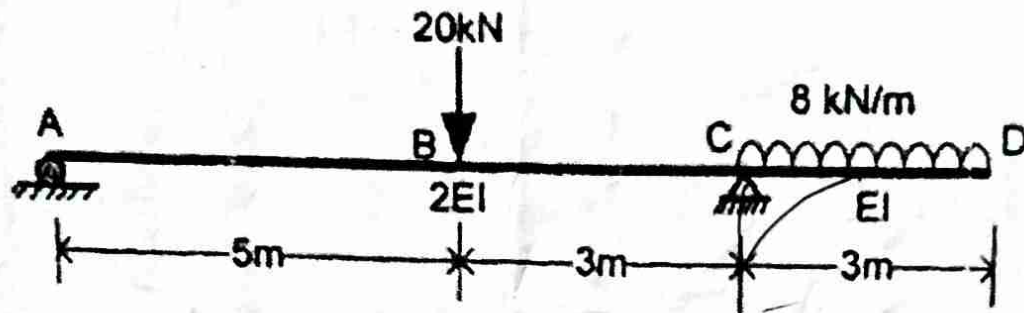


Fig. 3

By unit load method, determine the horizontal displacement of the joint G and vertical deflection of the joint J of the truss shown in the fig. 4. There is a temperature drop of 40°C in lower chord member only. Coefficient of expansion or contraction is $11.7 \times 10^{-6} / ^\circ\text{C}$. The C/S area of bottom and top chord members is 40000 mm^2 while that of the vertical and diagonal ties is 10000 mm^2 .

Take $E = 2 \times 10^5 \text{ N/mm}^2$.

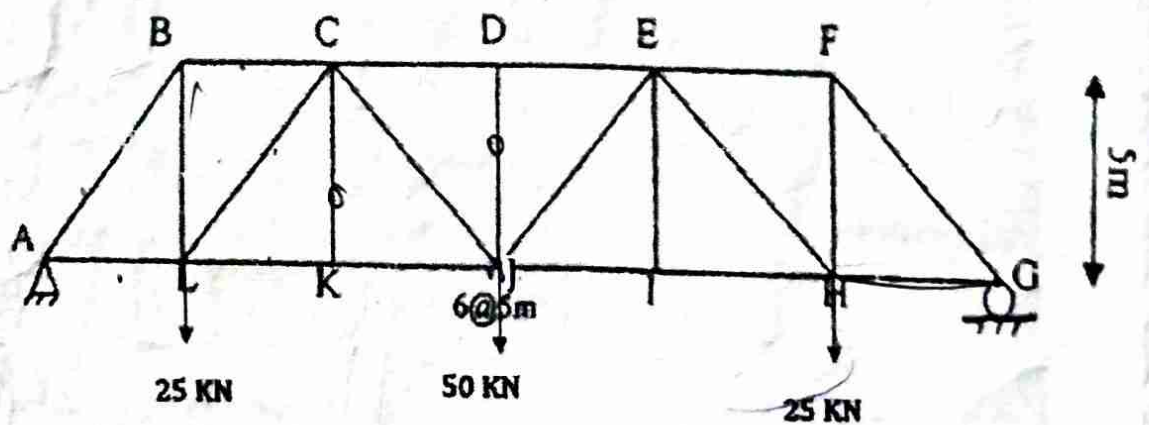
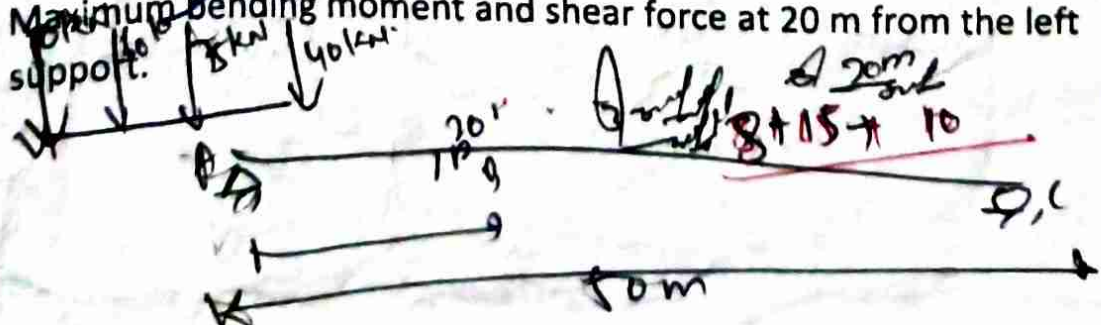


Fig. 4

9. A train of wheel load as shown in the figure 5 crosses a girder of 50 m span with 40 kN load leading. Determine the value of:

- Absolute Maximum bending moment of the girder.
- Maximum bending moment and shear force at 20 m from the left support.



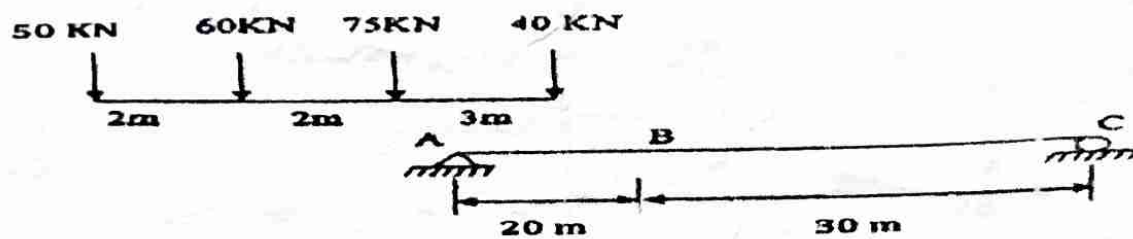


Fig. 5

10. (a). Draw an ILD for Bending moment and shear force at a section $L/4$ in the three hinged stiffening girder of span L . *S.H.*

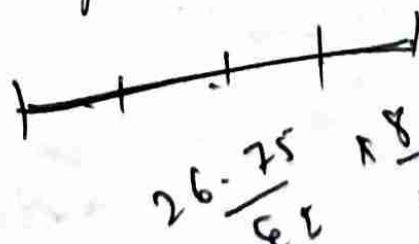
(b) A cable is designed to support a bridge over Malakai River. The river has a span of 150 m and the cable has a central dip of 10 m loaded with the 3 point loads of 25 kN each placed in equal interval in addition with the udl of intensity of 15 kN/m throughout the span. The supports are at the same level. Calculate the length of the cable, Maximum tension in the cable and also suggest the necessary diameter of the cable if the allowable stress in the cable is 12 N/mm^2 .

11. (a) Determine a position and magnitude of the maximum instantaneous deflection as a result of a falling weight of 10 kN from a height of 300 mm in a simply supported beam initially loaded with 5 kN/m. Consider span of beam = 5m, cross section = 200X 400 mm and $E = 200 \text{ GPa}$. Neglect the self-weight of the beam.

(b) Deduce and draw an influence line diagram for shear force and bending moment for both sides overhanging simply supported beam of span l at $l/4$ from the right support.

$$L = l_1 + \frac{l_1 h^2}{3}$$

length of cable =



30
24